

Do **not** write solutions on this page.

Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

10. [Maximum mark: 16]

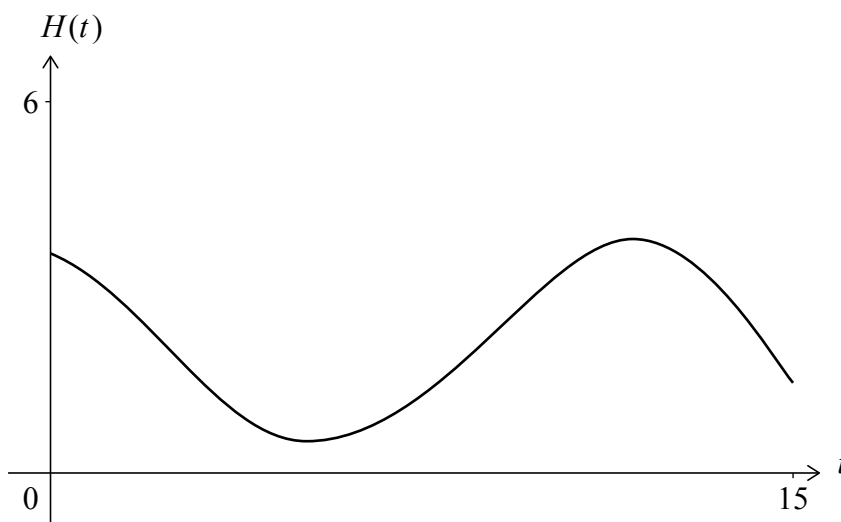
Sule Skerry and Rockall are small islands in the Atlantic Ocean, in the same time zone.

On a given day, the height of water in metres at Sule Skerry is modelled by the function $H(t) = 1.63 \sin(0.513(t - 8.20)) + 2.13$, where t is the number of hours after midnight.

The following graph shows the height of the water for 15 hours, starting at midnight.

At low tide the height of the water is 0.50 m. At high tide the height of the water is 3.76 m.

All heights are given correct to two decimal places.



- (a) The length of time between the first low tide and the first high tide is 6 hours and m minutes. Find the value of m to the nearest integer. [3]
- (b) Between two consecutive high tides, determine the length of time, in hours, for which the height of the water is less than 1 metre. [2]
- (c) Find the rate of change of the height of the water when $t = 13$, giving your answer in metres per hour. [2]

(This question continues on the following page)



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(Question 10 continued)

On the same day, the height of water at the second island, Rockall, is modelled by the function $h(t) = a \sin(b(t - c)) + d$, where t is the number of hours after midnight, and $a, b, c, d > 0$.

The first low tide occurs at 02:41 when the height of the water is 0.40 m.

The first high tide occurs at 09:02 when the height of the water is 2.74 m.

(d) Find the values of a, b, c and d . [7]

When $t = T$, the height of the water at Sule Skerry is the same as the height of the water at Rockall for the first time.

(e) Find the value of T . [2]

